LOURDES MATHA
COLLEGE OF SCIENCE & TECHNOLOGY
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Managed by Archdiocese of Changanacherry

Department of Computer Applications

Workbook

20MCA283-Deep Learning

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DEEP LEARNING (20MCA283)

MODULE-I

PART –A (3 marks each)

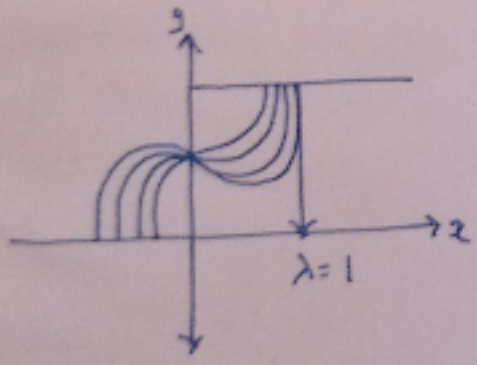
1. Describe sigmoid activation functions.

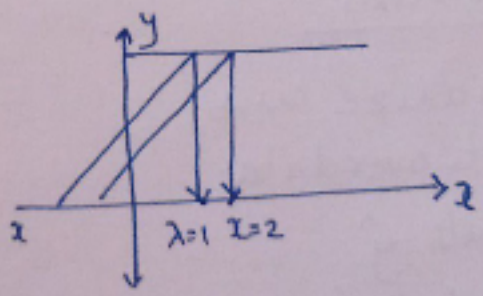
Sigmoidal function widely used in backpropagation ~~blw between~~ because of the relationship between the value of the function at a point and value of drivative at that point which reduce the computational burden during training. Two types are:

i) Binary sigmoid function
ii) Bipolar sigmoid function

i) Binary Sigmoid function : This activation function performs i/p editing blw 0 and 1. It is tve in nature. It is always bounded which means its o/p cannot be less than 0 and more than 1. It is also strictly increasing in nature which means more the i/p higher would be the o/p.

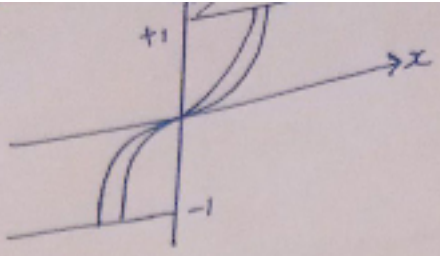
$$f(x) = Sigm(x) = \frac{1}{1+exp(-x)}$$





ii) Bipolar Sigmoidal function :- This activation function performs i/p editting b/w -1 and 1. It can be tve or -ve in nature. It is always bounded which means its o/p cannot be less than -1 and more than 1. It is also strictly increasing in nature like sigmoid function.

$$f(x) = Sigm(x) = \frac{2}{1+exp(-x)} - 1 = \frac{1-exp(x)}{}$$



2. Write the gradient descent algorithm.

Gradient descent algorithm : Iteratively calculate the next point using gradient of the current po Then scalar it by a learning rate and subtract th obtained values from the current position. 'η' eta learning rate it scalar the gradient and thus control the step size when eta become smaller t longer is gradient descent converges.

Steps

1. Choose a starting point (initialization)
2. Calculate gradient at this point
3. make a scaledstep in the opposite direction to the gra (minim
4. Repeat Step 2 & 3 until one of the criteria
5. maximum no: of iteration is reached
6. Step size is smaller than the tolerence

Algorithm

Initialize w, b;
iterate overdata;
compute $\hat{y}$
compute $L(w, b)$

$$w_{t+1} = w_t - \eta \Delta w_t$$

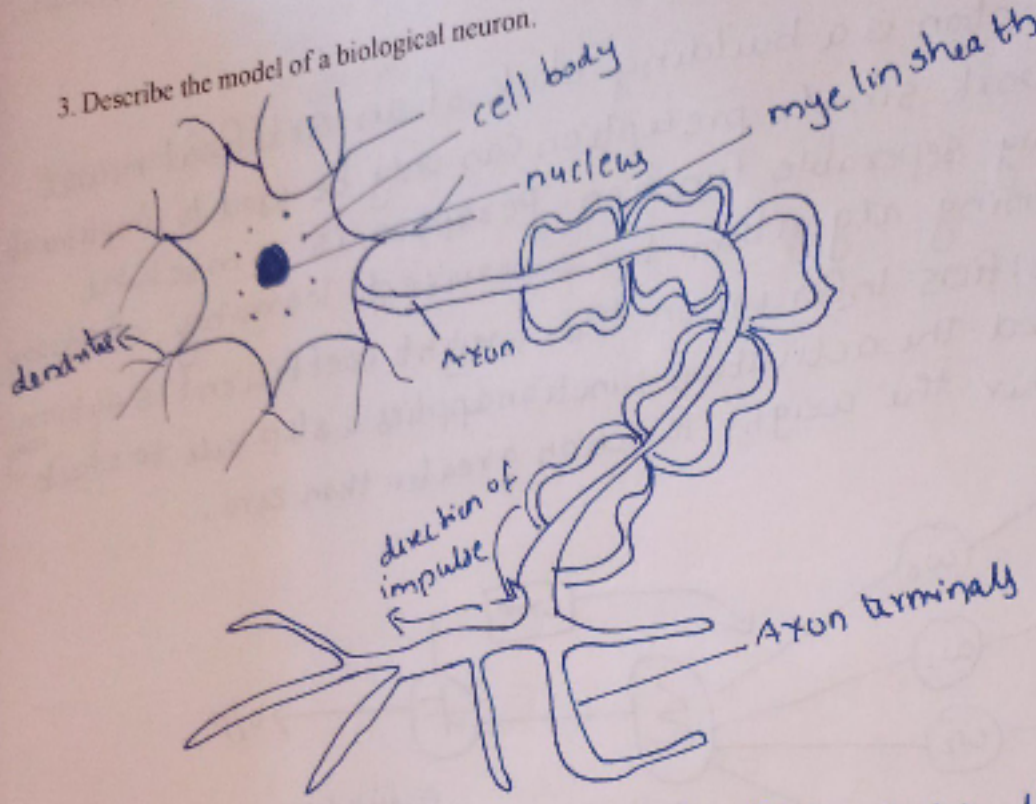
$$b_{t+1} = b_t - \eta \Delta b_t$$

where $\Delta w_t = \frac{\partial L(w, b)}{\partial w}$

at $w = w_t, b = b_t$

$\Delta b_t = \partial L(w, b)$

3. Describe the model of a biological neuron.



1. The majority of neurons encode their activation or o/p as a series of brief electrical pulses

2. The neuron's cell body (soma) processes the incoming activations and converts them into o/p activations.

3. The neuron's nucleus contains the genetic material in the form of DNA. This exists in most types of cells not just neurons.

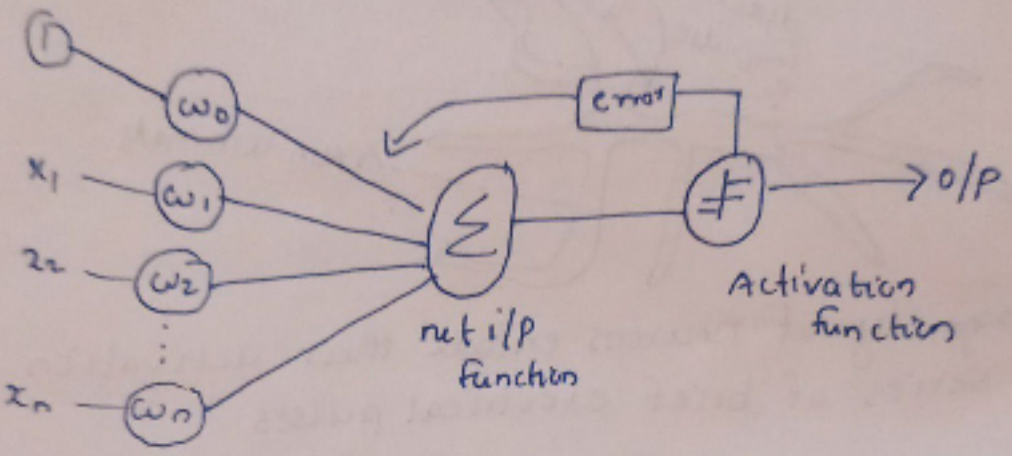
4. Dendrites are fibres which emanate from the cell body and provide the receptive zones the receive activation from other neurons.

5. Axons are fibres acting as trasmission lines that send activation to other neurons.

6. The Junctions that allow signal transmission blw the axon and dendrites are called synapse. The process of transmission is by diffussion of chemicals called neuro transmitters across the synaptic cleft.

4. Explain Perceptron learning algorithm.

Perceptron is a building block of an Artificial neural network. single perceptron can only be used to implement linearly seperable function. Perceptron is a machine Learning algorithm for supervised learning of binary classifiers. In Perceptron the weight coefficient is automatically learned. The activation functionapplies a step rule to check whether the weight function greater than zero.



step 1 $\Sigma w_i * x_i = x_1 * w_1 + x_2 * w_2 \cdots w_n * x_n$

$\Sigma w_i * x_i + b$

step 2 $Y = f(\Sigma w_i * x_i + b)$

Algorithm

P ← i/ps with label 1;

N ← i/ps with label 0;

initialize ω randomly;

while !(convergence do

Pick random $x \in P \cup N$;

if $x \in P$ and $\omega \cdot x < 0$ then

$\omega = \omega + x$;

end

if $x \in N$ and $\omega \cdot x \geq 0$ then

$\omega = \omega - x$;

end

end